Biology / Secondary

Diffusion and concentration gradients

Diffusion is the net movement of particles from higher to lower concentration as a result of their random motion.

Core ideas

1. A concentration gradient is a difference in concentration between two regions.

2. Particles move randomly in both directions. The net movement is down the gradient.

3. In comparable conditions, a larger surface area, steeper gradient or shorter diffusion distance can increase the rate of diffusion.

Worked example

If oxygen concentration is higher in an air space than in nearby blood, there can be net movement of oxygen towards the blood down its concentration gradient.

Try it yourself

1. State the direction of net diffusion.

2. Do particles stop moving when concentrations become equal?

3. How does reducing diffusion distance affect the rate in comparable conditions?

Check your answers

1. From higher to lower concentration.

2. No. Random movement continues, but there is no overall net movement.

3. It increases the rate.